LAB 3

Antenna CAD Software: CST Microwave Studio

Objective

The objective is to get acquainted with CAD software for antennas and particularly with *CST Microwave Studio.*

Pre- Work

None.

Lab-work

1. Follow the tutorial to design a microstrip patch antenna resonating at 2.45 GHz. Notice that you are using a new version of the software and some option may be a little different. In particular:
   - After creating the coaxial feeding the software detects the intersection between the coaxial and the ground plane. You will be prompted with a new “Shape intersection” menu. Choose “cut away highlighted shape”. This will remove the part of the ground plane that intersects with the coaxial feeding.
   - The solver you have to choose is the “Time domain solver”.

2. Once you have followed the tutorial instructions modify the patch geometry to obtain the resonant frequency of 2.45 GHz and an input impedance of 50Ω at that frequency.
   - The minimum return losses are obtained at frequency below 2.4 GHz. Use the marker functions (double click on the curve) to obtain that frequency.
   - Modify the dimension of the patch length to move that minimum to 2.45 GHz. HINT: you can use math expressions in the parameter list. The change is proportional to the ratio between frequencies.
   - Once you have obtained the desired frequency watch the S11 into the Smith Chart (the display options are on the top right). Obtain the value of the impedance at 2.45 GHz (double click on the curve and move the marker).
   - The real part of the impedance is not 50Ω. You can change it by moving the feeding point. Moving it to the center lowers the impedance, moving it outwards increases the impedance. Modify the variable CoaxPos to obtain a 50Ω impedance.
   - Estimate the antenna Q by measuring the 10 dB Return Loss fractional bandwidth.
3. Now double the height of the patch antenna and readjust it and obtain the Q. What can you say about the dependence of the Q and the patch height?